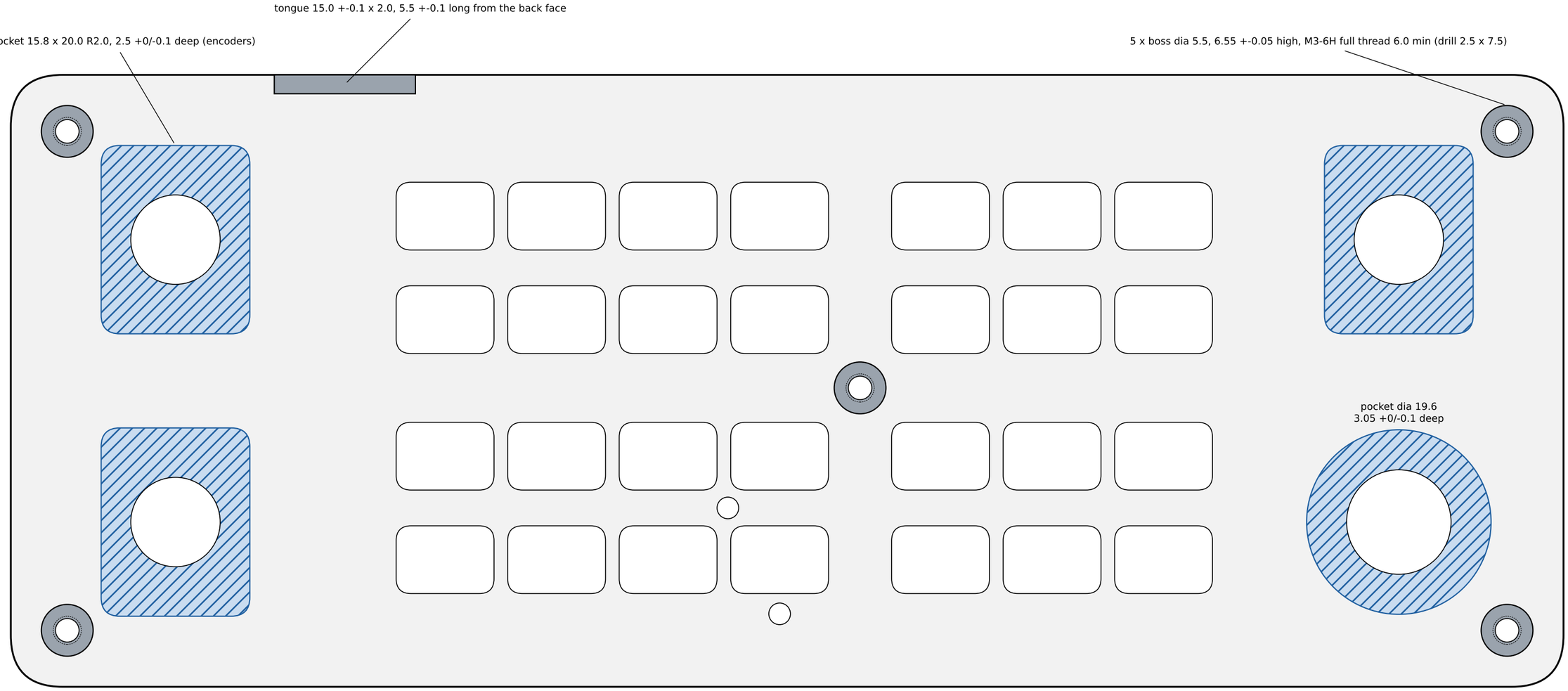
Avionics Panel rev 1 front plate, seen from the BACK (machined side)



Features

key hole	10.4 x 7.2 +0.1/-0 R1.6	through, corner R1.7 max	x28
encoder hole	dia 9.5	through	x3
encoder pocket	15.8 x 20.0 R2.0	2.5 +0/-0.1 deep from back, floor corner R0.3 max	x3
RANGE hole	dia 11.1	through	x1
RANGE pocket	dia 19.6	3.05 +0/-0.1 deep from back, floor corner R0.3 max	x1
SHIFT light hole	dia 2.3 +0.1/-0	through	x1
UNIT light hole	dia 2.3 +0.1/-0	through	x1
boss	dia 5.5, root R0.3 max	6.55 +-0.05 high from back, end face flat	x5
tapped hole	M3-6H in each boss	full thread 6.0 min, drill 2.5 x 7.5 deep	x5
tongue	15.0 +-0.1 x 2.0, root R0.3 max	5.5 +-0.1 long from back, outer face flush with plate edge	x1

Notes

- Material 6061 aluminium, machined from 12 mm plate: the web 4.0 +-0.1 mm thick before anodising, the 5 bosses and the tongue on its back, 10.55 mm high in all (the front may be faced). Outline 165.0 x 65.0 mm, corner R5.5. Volume 31.7 cm3. Units mm, general tolerance ISO 2768-m; flatness 0.1 over the whole web after bead blasting and anodising (the plate is held only at its bosses).
- Front face (+Z in the STEP) is flat: no pockets or fastener holes on the front, and no chamfers beyond the edge break below. Machining order: face the front first, then machine the back (pockets, bosses, tongue, holes), so the web is never faced while it is held on its bosses.
- Positions of the 5 bosses and their tapped holes, of the pockets, of the tongue and of all through holes: +-0.1 to each other (the key caps, the board, the parts in the pockets, the screws and the body's slot must line up).
- Bosses: dia 5.5 +-0.1, 6.55 +-0.05 high from the back face of the web (required: they set the plate height), each end face flat and square to its hole; root R0.3 max.
- Tongue on the back edge: 15.0 +-0.1 wide, 2.0 thick with its outer face flush with the plate's edge, 5.5 +-0.1 long from the back face of the web (it ends short of the bosses); root R0.3 max. It closes the body's slot over the USB-C socket.
- Threads: 5 x M3-6H, one in each boss from its end face, full thread at least 6.0 deep, tap drill dia 2.5 x 7.5 deep plus the drill point, 0.3 x 90 deg countersink: at least 2.15 mm of metal stays in front of the drill point. The threads must gauge M3-6H after anodising: mask them while anodising, or cut them again afterwards. The boss end faces may be anodised or bare, but all of them alike.
- Pocket depths +0, never deeper: at least 0.85 mm of material must stay to the front face under every pocket; nothing may break through or mark the front face.
- Encoder pockets and RANGE pocket: floor corner (inside radius between wall and floor) at most 0.3 mm; the parts' top corners come within about 0.2 mm of the pocket walls.
- Finish: black anodised, matte (bead blasted before anodising); the pockets may be anodised or bare, but all pockets alike (all anodised or all bare). Anodising rack contacts in one of the M3 threads (cut it again afterwards), on the back face, on the boss end faces or inside the pockets; never on the front face, the outline or the key holes.
- Marking on the front per avionics-panel-rev1-plate.dxf, same origin (bottom-left, seen from the front): layer LASER = laser marking, layer UV_AMBER = UV print in JLCCNC's standard orange. Layer CUT = outline and through holes for reference. DXF units mm; fill closed contours even-odd (letter counters are separate contours).
- Key holes 10.4 x 7.2 +0.1/-0 after anodising (never smaller: the key caps centre the board in them), corners R1.6, R1.7 max. Light holes +0.1/-0 (never smaller: the light pipes slide in them). Deburr all edges: edge break at most 0.2 on the front outline and hole edges; on the back edges of the key holes 0.3 to 0.5 (a lead-in for the key caps); the light holes' back edges sharp, deburred only (0.05 max: the light pipes' shoulders bear there).